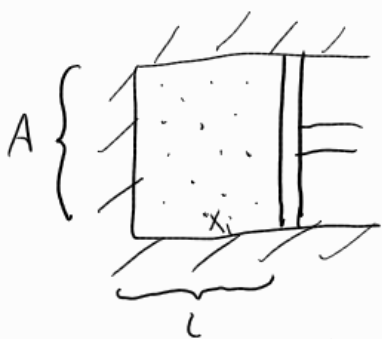


$$U_i(x_i) = \begin{cases} 0, & x_i < l \\ \infty, & x_i > l \end{cases}$$

$$U_i(x_i, l) = U_i(x_i - l)$$

Sila na bat

$$F_x = - \sum_i \frac{\partial U_i}{\partial x_i} = + \sum_i \frac{\partial U_i}{\partial l} = \frac{\partial}{\partial l} \sum U_i = \frac{\partial U}{\partial l} = \frac{\partial \epsilon}{\partial l} \quad \text{ker } T \text{ ni odv. od } l$$

$$p = - \frac{\overline{F_x}}{A} = - \frac{1}{A} \frac{\partial \epsilon}{\partial l} = - \frac{\partial \epsilon}{\partial (Al)} = - \frac{\partial \epsilon}{\partial V}$$

Fazna vsota:

$$e^{-\beta F} = c \int \exp(-\beta \epsilon) d\Gamma$$

$$1 = c \int e^{-\beta(\epsilon - F)} d\Gamma \quad / \frac{\partial}{\partial V}$$

$$0 = c \int \left(\frac{\partial \epsilon}{\partial V} - \frac{\partial F}{\partial V} \right) (-\beta) \exp(-\beta(\epsilon - F)) d\Gamma = -\beta c \left(\int \frac{\partial \epsilon}{\partial V} \exp(\dots) d\Gamma - \int \frac{\partial F}{\partial V} \exp(\dots) d\Gamma \right)$$

$$\underbrace{c \int \frac{\partial \epsilon}{\partial V} \exp(-\beta(\epsilon - F)) d\Gamma}_{\frac{\partial \epsilon}{\partial V}} = \frac{\partial F}{\partial V} \underbrace{c \int \exp(-\beta(\epsilon - F)) d\Gamma}_1$$

$$\rightarrow \boxed{p = - \frac{\partial \epsilon}{\partial V} = - \left(\frac{\partial F}{\partial V} \right) / \beta}$$

Zgled: idealni plin

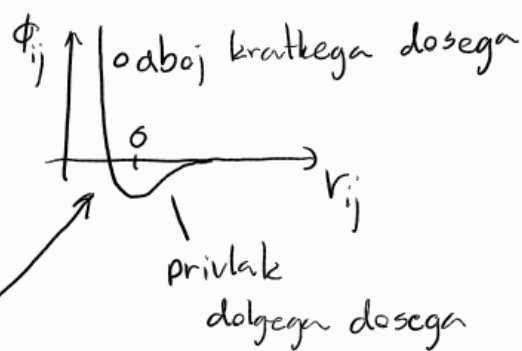
$$e^{-\beta F} = c V^N \left(\frac{2\pi m}{\beta} \right)^{3N/2}$$

$$\beta F = -\ln c - N \ln V - \frac{3N}{2} \ln \frac{2\pi m}{\beta}$$

$$p = - \frac{\partial F}{\partial V} = \frac{N}{\beta V} = \frac{N k_B T}{V}$$

Realni plin

meddelčni potencial $\phi_{ij} = \phi(r_{ij})$



Lennard-Jonesov 12-6 potencial

$$\phi(r_{ij}) = \epsilon \left[\left(\frac{r_{ij}}{\sigma} \right)^{12} - 2 \left(\frac{r_{ij}}{\sigma} \right)^6 \right]$$

parne interakcije

$$U = \sum_{i < j} \phi_{ij} = \phi_{12} + \phi_{13} + \dots + \phi_{1N} + \phi_{23} + \dots + \phi_{3N} + \dots$$

$$e^{-\beta F} = C \int \exp(-\beta(T+U)) d\vec{r}_1 \dots d\vec{r}_N d\vec{p}_1 \dots d\vec{p}_N = C \left(\frac{2\pi m}{\beta} \right)^{3N/2} \underbrace{\int \exp(-\beta \sum_{i < j} \phi_{ij}) d\vec{r}_1 \dots d\vec{r}_N}_{Z}$$

$\xrightarrow{T(\vec{p}_1, \dots, \vec{p}_N)}$

$$Z = \int \exp(-\beta \dots) d\vec{r}_2 \dots d\vec{r}_N \underbrace{\exp(-\beta(\phi_{12} + \phi_{23} + \dots))}_{\text{exp}(-\beta\phi_{12}) \text{exp} \dots} d\vec{r}_1$$

$$V - (N-2)2B_1 \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\exp(-\beta\phi_{12}) \exp \dots$$

$$[1 - (1 - \exp(-\beta\phi_{12}))] \dots$$

$$\approx 1 - [1 - \exp(-\beta\phi_{12})] - [1 - \exp(-\beta\phi_{13})] - \dots$$

$$\int d\vec{r}_1 \quad \downarrow \quad \downarrow$$

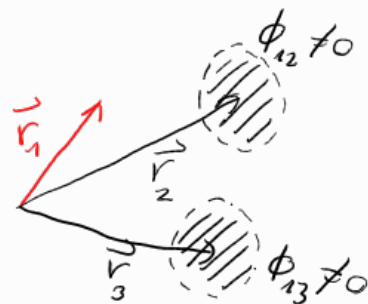
$$V - \underbrace{\int [1 - \exp(-\beta\phi_{12})] 4\pi r_{12}^2 dr_{12} - \int [1 - \exp(-\beta\phi_{13})] 4\pi r_{13}^2 dr_{13} - \dots}_{N-1 \text{ členov}}$$

$$V - (N-1)2B_1; \quad 2B_1 = \int \phi 4\pi r^2 dr$$

drugi virialni koef.

$$\Rightarrow e^{-\beta F} = C \left(\frac{2\pi m}{\beta} \right)^{3N/2} [V - (N-1)2B_1] [V - (N-2)2B_1] \dots V \approx$$

$$= C \left(\frac{2\pi m}{\beta} \right)^{3N/2} V^N \left(1 - (N-1) \frac{2B_1}{V} \right) \left(1 - (N-2) \frac{2B_1}{V} \right) \dots \approx e^{-\beta B_1} \ll V$$



$$\approx C \left(\frac{2\pi m}{\beta} \right)^{3N/2} V^N \left[1 - \frac{B_1}{V} \underbrace{((N-1) + (N-2) + \dots + 1)}_{\frac{(N-1)N}{2} \approx N^2} \right]$$

$$e^{-\beta F} = \underbrace{C \left(\frac{2\pi m}{\beta} \right)^{3N/2}}_{\text{idealni plin}} \underbrace{V^N \left[1 - N^2 \frac{B_1}{V} \right]}_{\text{odstopanje od idealnega plina}}$$

$$F = -\frac{1}{\beta} \left(\ln C + \frac{3N}{2} \ln \frac{2\pi m}{\beta} + N \ln V + \ln \left(1 - \underbrace{\frac{N^2 B_1}{V}}_{\approx \frac{N^2 B_1}{V}} \right) \right)$$

$$p = -\frac{\partial F}{\partial V} = \frac{1}{\beta} \left(\frac{N}{V} + \frac{N^2}{V^2} B_1 \right) =$$

$$= \boxed{\frac{N k_B T}{V} \left(1 + \frac{N B_1}{V} \right)}$$

drugi virialni koef.:

pribl.: $\Phi(r) = \begin{cases} \infty; & r < \sigma \\ -\phi_p; & r > \sigma \end{cases}$

$$B_1 = \frac{1}{2} \left[\int_0^\sigma (1 - \exp(-\beta \infty)) 4\pi r^2 dr + \int_\sigma^\infty [1 - \exp(-\beta \phi_p)] 4\pi r^2 dr \right] =$$

$$= \frac{1}{2} \left[\underbrace{\frac{4\pi}{3} \sigma^3}_{8V_0} - \underbrace{\beta \int_\sigma^\infty \phi_p 4\pi r^2 dr}_{2\alpha; \alpha > 0} \right] = 4V_0 - \frac{\alpha}{k_B T}$$

$$\rightarrow p = \frac{N k_B T}{V} \left(1 + \frac{N}{V} \left(4V_0 - \frac{\alpha}{k_B T} \right) \right) = \frac{N k_B T}{V} \left(1 + \frac{4NV_0}{V} \right) - \frac{\alpha N^2}{V^2}$$

VDW:

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2} \stackrel{V_m \gg b}{\approx} \frac{RT}{V_m} \left(1 + \frac{b}{V_m} \right) - \frac{a}{V_m^2} \rightarrow \begin{matrix} b \leftrightarrow 4V_0 \\ a \leftrightarrow \alpha \end{matrix}$$